Introduction into Possibilistic Logic

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Overview

What are Possibility Distributions?

- A **possibility distribution** is a mapping:

$$\pi : U \to S,$$

  where:
  - U: Set of states (domain of an attribute, Cartesian product, propositional interpretations, etc.).
  - S: Totally ordered scale ($S = [0, 1]$ or finite chains).
  - $\pi(u)$: Degree of possibility (plausibility) of state u.
- Top of the scale: 1 (fully plausible).
  Bottom of the scale: 0 (impossible).

Interpretation of Possibility Distributions

- $\pi(u) = 0$: State u is **impossible**.
- $\pi(u) = 1$: State u is **fully plausible**.
- Higher $\pi(u)$ values mean greater plausibility.

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Normalization Condition

At least one state $u \in U$ must have $\pi(u) = 1$ (an actual state exists).

Overview

Specificity of a Possibility Distribution

- A distribution π is **at least as specific** as π' if:

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

- Examples:
  - **Complete knowledge:** $\pi(u_0) = 1$, $\pi(u) = 0$ for all $u \neq u_0$.
  - **Complete ignorance:** $\pi(u) = 1$ for all $u \in U$.

The Principle of Minimal Specificity

Definition

Any hypothesis not known to be impossible cannot be ruled out. Possibility degrees should be maximized under given constraints.

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- If x is restricted by fuzzy set F ($\mu_F(x)$), then:

$$\pi(x) \leq \mu_F(x).$$

- Combining multiple pieces of knowledge:

$$\pi(x) = \min_{i=1}^{n} \mu_{F_i}(x).$$

Overview

- Possibility measure:

$$\Pi(A) = \sup_{u \in A} \pi(u).$$

  Evaluates the extent to which A is consistent with π.

- Necessity measure:

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

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Duality Relationship

$$N(A) = 1 - \Pi(A^c).$$

Where A^c is the complement of A.

- **Possibility: Maxitivity Property**

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)).$$

- **Necessity: Minimization Property**

$$N(A \cap B) = \min(N(A), N(B)).$$

Overview

Example: Knowledge Scenarios

- $U = \{u_1, u_2, u_3\}$, and $\pi(u_1) = 0.8, \pi(u_2) = 0.6, \pi(u_3) = 0.2$.
- For $A = \{u_1, u_2\}$:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(u_1), \pi(u_2)) = 0.8.$$
$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(u_3) = 0.8.$$

Overview

Applications of Possibility Theory

- **Default Reasoning**: Handling incomplete knowledge in expert systems.
- **Medical Diagnosis**: Prioritizing plausible diagnoses based on symptoms.
- **Decision-Making**: Evaluating options under uncertainty.
- **Fuzzy Control Systems**: Representing and processing vague information.

Overview

What is Possibilistic Logic?

- **Definition**: Possibilistic logic is an extension of classical logic designed to handle:
  - **Uncertainty**: Representing incomplete knowledge.
  - **Plausibility**: Differentiating between plausible and less plausible states.
- **Core Measures**:
  - **Possibility (Π)**: Indicates how plausible an event is.
  - **Necessity (N)**: Indicates how certain an event is.
- **Applications**:
  - Used in expert systems, decision-making, and AI for reasoning under uncertainty.

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Key Idea

Possibilistic logic prioritizes **qualitative reasoning** over quantitative probabilities, making it useful for handling incomplete or vague information.

Overview

Syntax of Possibilistic Logic

- A possibilistic logic formula is represented as:

$$(a, \alpha)$$

  where:
  - a: A classical logic formula (e.g., $p \to q$).
  - $\alpha \in [0, 1]$: Certainty level (minimum necessity degree of a).
- **Interpretation**:
  - (a, α): "Formula a is at least certain to degree α."
  - The higher α, the stronger the belief in a.

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Example

Rule: *"If fever, then flu" is 90% certain.*

$$(\text{fever} \to \text{flu}, 0.9)$$

- This means: "If fever is observed, it is 90% certain that the patient has flu."

Overview

- **Possibility Measure (Π)**:

$$\Pi(A) = \sup_{u \in A} \pi(u)$$

  - Evaluates the highest degree of plausibility in set A.
- **Necessity Measure (N)**:

$$N(A) = \inf_{u \notin A} (1 - \pi(u))$$

  - Evaluates how strongly A is implied by the information.
- **Duality**:

$$N(A) = 1 - \Pi(A^c)$$

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Example (Diagnoses)

Let $U = \{\text{flu}, \text{cold}, \text{allergy}\}$ and:

$$\pi(\text{flu}) = 0.9, \pi(\text{cold}) = 0.6, \pi(\text{allergy}) = 0.2$$

Overview

Inference Rules

- **Weakened Certainty**:

$$(a, \alpha) \Rightarrow (a, \beta), \quad \text{if } \alpha \geq \beta$$

- **Min-Based Conjunction**:

$$N(a \land b) = \min(N(a), N(b))$$

- **Resolution**:

$$(\neg a \lor b, \alpha), (a \lor c, \beta) \Rightarrow (b \lor c, \min(\alpha, \beta))$$

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Example (Continued)

Using rules:

$$(\neg \text{fever} \vee \text{flu}, 0.9), \quad (\text{fever} \vee \text{headache}, 0.8)$$

Inference:

$$(\text{flu} \vee \text{headache}, 0.8)$$

Overview

- Possibilistic logic can handle conflicting information by:
  - Identifying an **inconsistency level** ($inc-l$).
  - Filtering out rules with certainty levels below $inc-l$.
- **Example**:

$$\{(a, 0.8), (\neg a, 0.7)\}$$

  - Inconsistency level: $inc - l = 0.7$.
  - Filtered result: $(a, 0.8)$ dominates.

Overview

Applications of Possibilistic Logic

- **Medical Diagnosis**:
  - Rules with certainty for differential diagnoses.
- **Decision-Making**:
  - Selecting actions under incomplete knowledge.
- **Knowledge Bases**:
  - Handling updates with belief revision.

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Example Wrap-Up

Throughout the presentation, we used:

- Syntax: Representation of rules $((a, \alpha))$.
- Measures: Calculating $\Pi(A)$ and $N(A)$.
- Inference: Combining uncertain rules.
- Inconsistencies: Managing conflicts in rules.

Overview

What is Possibilistic Logic?

- **Definition**: Possibilistic logic is an extension of classical logic that incorporates a certainty level for each formula.
- **Purpose**: Designed for reasoning under qualitative uncertainty and preferences.
- **Core Characteristics**:
  - A formula in possibilistic logic is a pair (a, α), where:
    - a: A classical logical formula (propositional or first-order logic).
    - $\alpha \in [0, 1]$: A certainty level.
  - Certainty levels (α) represent the confidence or priority of the formula.
- **Scope**:
  - Classical logic is retained: Formulas are either true or false.
  - Extensions exist for fuzzy predicates, but basic possibilistic logic does not handle degrees of truth.

Overview

Basic Syntax

- A possibilistic logic formula is written as:

$$(a, \alpha), \quad \text{where } \alpha \in [0, 1].$$

- Interpretation:
  - a: A propositional formula.
  - α: The minimum necessity level associated with a.
- **Examples**:
  - $(p \to q, 0.8)$: "If p, then q with certainty 0.8."
  - $(\neg p, 1)$: "Not p is absolutely certain."

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Key Idea

Possibilistic logic extends classical logic by associating a priority (certainty) level to each formula while maintaining Boolean semantics.

Overview

Inference Rules

- **Certainty Weakening**:

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta).$$

- **Modus Ponens**:

$$(\neg a \vee b, \alpha), \quad (a, \alpha) \vdash (b, \alpha).$$

- **Resolution**:

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha).$$

- **Weakest Link Resolution**:

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)).$$

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Example

Consider:

$$(\neg \text{fever} \vee \text{flu}, 0.9),$$
$$(\text{fever} \vee \text{headache}, 0.8).$$

Formula Weakening

- **Definition**:

$$\text{If } a \vdash b \text{ in classical logic, then } (a, \alpha) \vdash (b, \alpha).$$

- Derived from resolution:

$$(\neg a \vee b, 1), \quad (a, \alpha) \vdash (b, \alpha).$$

- Example:
  - $a \rightarrow b$: $p \wedge q \vdash p$.
  - With certainty: $(p \wedge q, 0.7) \vdash (p, 0.7)$.

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Key Property

Any valid deduction in propositional logic is valid in possibilistic logic, with certainty levels propagated.

- A possibilistic logic base is a collection of weighted formulas:

$$\{(a_1, \alpha_1), (a_2, \alpha_2), \dots\}.$$

- Clauses can be rewritten in terms of their weights:
  - Example:

$$(a \wedge b, \alpha) \quad \Leftrightarrow \quad \{(a, \alpha), (b, \alpha)\}.$$

- Logical tautologies (t) hold with any certainty level:

$$(t, \alpha), \quad \forall \alpha \in [0, 1].$$

Overview

Example: Medical Diagnosis

- Rule base:

$$(\neg\text{fever} \vee \text{flu}, 0.9),$$
$$(\neg\text{cough} \vee \text{cold}, 0.8),$$
$$(\text{fever} \vee \text{headache}, 0.7).$$

- Facts:

$$(\text{fever}, 1), \quad (\text{cough}, 1).$$

- Inference:
  - From Modus Ponens:

$$(\text{flu}, 0.9), \quad (\text{cold}, 0.8).$$

  - From Resolution:

$$(\text{flu} \vee \text{headache}, 0.7).$$

Overview

Conclusion

- Possibilistic logic provides a robust framework for reasoning under uncertainty.
- Key features:
  - Certainty levels for qualitative inference.
  - Compatibility with classical logic.
  - Handling of conflicting information.
- Applications:
  - Medical diagnosis, decision-making, and knowledge representation.

Overview

What is Possibilistic Logic?

- **Definition**: A logical framework for reasoning under uncertainty.
- **Purpose**: Handles incomplete information and default rules with exceptions.
- **Core Idea**:
  - Classical logic enriched with certainty levels.
  - Focus on **default reasoning**: "What typically happens."
- **Applications**:
  - Nonmonotonic reasoning.
  - Knowledge representation and reasoning about exceptions.

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Example Context

Default reasoning can handle statements like:

- "Birds fly (usually)." ($b \rightarrow f$).
- "Penguins do not fly." ($p \rightarrow \neg f$).
- "Penguins are birds." ($p \rightarrow b$).

Overview

- A possibilistic formula is a pair:

$$(a, \alpha), \quad \text{where } \alpha \in [0, 1].$$

- Interpretation:
  - a: A propositional formula.
  - α: Certainty level (degree of belief in a).
- **Examples**:
  - $(b \rightarrow f, 0.9)$: Birds fly with certainty 0.9.
  - $(p \rightarrow \neg f, 1)$: Penguins do not fly, absolutely certain.

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Key Idea

Certainty levels enrich classical logic, enabling reasoning about typical cases and exceptions.

Overview

Key Inference Rules

- **Certainty Weakening**:

$$\underline{\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta)}.$$

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Penguin Example

Consider:

$$(\neg b \vee f, 0.9), \quad (\neg p \vee \neg f, 1), \quad (\neg p \vee b, 1).$$

Overview

Default Rules as Constraints

- Default rule:

$$\underline{\Pi(a \wedge b) > \Pi(a \wedge \neg b)}.$$

  - $a \rightarrow b$: "If a, then b is more plausible than $\neg b$."
- In terms of necessity:

$$\underline{N(b|a) = 1 - \Pi(\neg b|a) > 0}.$$

- Comparative relation:

$$\underline{a \wedge b >_{\Pi} a \wedge \neg b}.$$

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Penguin Example Constraints

$$\begin{aligned} b \wedge f &>_{\Pi} b \wedge \neg f && \text{(birds fly)}. \\ p \wedge \neg f &>_{\Pi} p \wedge f && \text{(penguins don't fly)}. \\ p \wedge b &>_{\Pi} p \wedge \neg b && \text{(penguins are birds)}. \end{aligned}$$

Overview

Computing Necessity for Penguin Example

- **Possibility Distribution**:

$$\pi(\omega) \text{ values assigned to worlds } \omega.$$

- Worlds:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

Computing Necessity for Penguin Example

- **Possibility Distribution**:

$$\pi(\omega) \text{ values assigned to worlds } \omega.$$

- Worlds:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

- **Constraints**:

$$C_1 : \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)).$$
$$C_2 : \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)).$$
$$C_3 : \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).$$

- Factual knowledge:

$$(b, 1) \quad \text{(Tweety is a bird).}$$

- Possibilistic inference:

$$K \cup \{(b, 1)\} \vdash (f, 1/3).$$

  - Default rule: Tweety flies with certainty 1/3.
- Adding $p = 1$ (Tweety is a penguin):

$$K \cup F \vdash (\neg f, 2/3).$$

  - New inference: Tweety does not fly.

Overview

Conclusion

- Possibilistic logic effectively models:
  - Default reasoning with exceptions.
  - Reasoning under uncertainty.
- **Penguin Example**:
  - Illustrated how rules like "birds fly" and "penguins don't fly" can be reconciled.
  - Demonstrated inference and conflict resolution.
- **Applications**:
  - Knowledge bases.
  - Default rule modeling.
  - Nonmonotonic reasoning.

Overview

Overview

Possibility distribution:

U is a set of affairs.

A possibility distribution is a mapping π from U to a totally ordered scale S, with top denoted by 1 and bottom by 0. In the finite case $S = \{1 = \lambda_1 > ...\lambda_n > \lambda_{n+1} = 0\}$.

The function π represents the state of knowledge of an agent (about the actual state of affairs), also called an epistemic state.

$\pi(u) = 0$ means that state u is rejected as impossible;

$\pi(u) = 1$ means that state u is totally possible ($=$ plausible).

Specificity: A possibility distribution π is said to be at least as specific as another π' if and only if for each state of affairs u: $\pi(u) \leq \pi'(u)$. Then, π is at least as restrictive and informative as π' , since it rules out at least as many states with at least as much strength.

Complete knowledge: for some u_0, $\pi(u_0) = 1$ and $\pi(u) = 0$, $\forall u \neq u_0$ (only u_0, is possible).

Complete ignorance: $\pi(u) = 1, \forall u \in U$ (all states are possible).

Principle of minimal specificity: It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

Overview

Overview

Blocks

Block

This is a regular block.

- This is an item in a block.

Block

This is an alert block.

- This is an item in an alert block.

Block

This is an example block.

- This is an item in an example block.

Overview

Highlighting Text

We can highlight text. There are also a darker option, and a yellow option.

The same options are also available in math mode: $a + b = c$

Overview

Appendix

This is an appendix.

- Page numbers start from the beginning.